

Small Unmanned Aerial Systems I

Center for Advanced Technologies (CAT) — Lakewood High School, Pinellas County

FAA Part 107 | Drone Operations | STEM Magnet Program

Course Overview

The sUAS 1 course is part of Lakewood High School's premier STEM magnet program, designed to equip students with the foundational knowledge and practical skills required to operate drones safely and professionally. The curriculum bridges regulatory compliance, aeronautical science, and real-world flight operations.

Core Curriculum

FAA Part 107 Exam Preparation

Comprehensive study of the Remote Pilot Certificate requirements — the legal standard for commercial drone operation in the United States.

Airspace & Regulations

In-depth coverage of airspace classifications, aeronautical chart interpretation, NOTAM procedures, and FAA operational rules.

Meteorology & Flight Planning

Weather pattern analysis, wind effects, and preflight planning for safe UAS operations.

Flight Operations & Emergency Procedures

Hands-on maneuvering in controlled environments, simulated real-world scenarios including aerial mapping and photography, and emergency response protocols.

Aeronautical Decision-Making

Risk assessment frameworks and crew resource management adapted for single-pilot UAS operations.

Hands-On Experience

- Assembled, programmed, and operated autonomous drone vehicles in team-based settings — including RC trucks, tracked rovers, quadcopters, and fixed-wing aircraft.
- Modified flight parameters and programmed autonomous missions; configured and operated ground-based radio controllers.
- Conducted systematic testing and performance evaluation of 3D-printed components and motors sourced from multiple vendors.
- Completed a structured Part 107 certification study plan covering vehicle requirements, airspace classification, aeronautical chart reading, and safety procedures.
- Completed comprehensive sUAS flight training and safety certification program
- Earned FAA Part 107 Remote Pilot-Federal Aviation Administration-Licensed for Small Unmanned Aircraft Systems (sUAS)

